

| Question |     |       |  | Marking details                                                                                                                   | Marks Available |          |          |           |          |          |
|----------|-----|-------|--|-----------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |  |                                                                                                                                   | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 6        | (a) | (i)   |  | Substitution: $W = \frac{1}{2} \times 0.15 \times 12$ (1)<br>= 0.9 [J] (1)                                                        | 1               | 1        |          | 2         | 2        |          |
|          |     | (ii)  |  | Substitution: 0.9 ( <b>ecf</b> ) = $0.5 \times 0.075 \times v^2$ (1)<br>Rearrangement: $v^2 = 24$ (1)<br>$v = 4.9$ (1)<br>m/s (1) | 1<br>1          | 1<br>1   |          | 4         | 3        |          |
|          |     | (iii) |  | Because some energy is lost (1)<br>To {heat / friction / resistive forces} (1)                                                    | 1               | 1        |          | 2         |          |          |
|          | (b) |       |  | Substitution: $0.8 = F \times 1.5$ (1)<br>$F = \frac{0.8}{1.5}$ (1)<br>$F = 0.5[3]$ [N] (1)                                       | 1               | 1<br>1   |          | 3         | 3        |          |
|          | (c) |       |  | [For the same extension] twice the force will be required (1)<br>So $\frac{1}{2} F_x$ will be double [so agree] (1)               |                 |          | 2        | 2         |          |          |
|          |     |       |  | <b>Question 6 total</b>                                                                                                           | <b>5</b>        | <b>6</b> | <b>2</b> | <b>13</b> | <b>8</b> | <b>0</b> |